

Active Suspension Design

Senior-level controls project | ME 4030 | Sriram Krishnamoorthy

Design and evaluate PID and LQR controllers for a quarter-car suspension. This corrected public edition preserves the original project scope and grading weights while clarifying signs, states, and performance measures.

System and assumptions

Model small vertical motions about static equilibrium with linear springs and dampers. The road displacement z_r is a disturbance. The actuator applies $+u$ to the body and $-u$ to the wheel. Assume an ideal force actuator; discuss the limits of this assumption.

Parameter	Value	Meaning
M_s	2500 kg	Sprung (body) mass
M_u	320 kg	Unsprung (wheel assembly) mass
K_s	80000 N/m	Suspension stiffness
C_s	350 N s/m	Suspension damping
K_t	500000 N/m	Tire stiffness
C_t	15020 N s/m	Tire damping

Equations of motion

$$\begin{aligned} M_s \ddot{z}_s &= -K_s(z_s - z_u) - C_s(\dot{z}_s - \dot{z}_u) + u \\ M_u \ddot{z}_u &= K_s(z_s - z_u) + C_s(\dot{z}_s - \dot{z}_u) \\ &\quad - K_t(z_u - z_r) - C_t(\dot{z}_u - \dot{z}_r) - u \end{aligned}$$

Use suspension deflection $s = z_s - z_u$ and tire deflection $h = z_u - z_r$. Tire forces are restoring: their signs in the wheel equation are negative for positive h and $\dot{h}$.

Road inputs and initial conditions

Use zero initial conditions and a 10 s record. The step road rises by 0.2 m at $t = 1$ s. The sinusoidal road is $z_r(t) = 0.1 \sin(2\pi \cdot 0.5 t)$ m. Report the simulation method and output time step. A sharp step with tire damping and ideal derivative feedback can create a very fast transient.

Performance measures

Evaluate tradeoffs; do not assume that every benchmark can be met simultaneously.

Compute RMS values over the full 10 s record. Let $\text{RMS}(v) = \sqrt{(1/T) \int_0^T v(t)^2 dt}$. Body acceleration is absolute vertical acceleration, not acceleration relative to the road.

Metric	Excellent	Good	Acceptable
Body acceleration RMS	$\leq 0.35 \text{ m/s}^2$	$\leq 0.50 \text{ m/s}^2$	$\leq 0.70 \text{ m/s}^2$
Tire deflection RMS	$\leq 0.008 \text{ m}$	$\leq 0.012 \text{ m}$	$\leq 0.020 \text{ m}$
Peak $ s $	$\leq 0.015 \text{ m}$	$\leq 0.025 \text{ m}$	$\leq 0.040 \text{ m}$
Actuator force RMS	$\leq 1.5 \text{ kN}$	$\leq 2.5 \text{ kN}$	$\leq 3.5 \text{ kN}$
Suspension recovery time	$\leq 3 \text{ s}$	$\leq 5 \text{ s}$	$\leq 7 \text{ s}$

Step-transient definitions

Suspension recovery: the first time after the road step beyond which $|s|$ remains at or below 0.004 m through the end of the simulation. Measure time from $t = 1 \text{ s}$. This is an absolute band equal to 2% of the road-step height. If the trajectory has not recovered by 10 s, report "not settled within the record".

Percentage overshoot of s : do not report it relative to a zero final value; that normalization is undefined. Report peak $|s|$ and recovery instead.

Supplemental body metrics: report body overshoot as $100 \max(0, \max(zs) - 0.2) / 0.2$ and body settling within $0.2 \pm 0.004 \text{ m}$. These are body-motion metrics, not suspension-deflection metrics. Do not apply step overshoot or settling definitions to the sinusoidal run.

Comparison table

For passive, PID, and LQR cases, report the five benchmark metrics above, peak actuator force, supplemental body metrics, controller parameters, and which benchmarks were met. Mark failed targets explicitly. These are instructional benchmarks, not certified vehicle comfort or safety standards.

Numerical accuracy

A 1 ms output grid is convenient for plots but may miss fast energy contributions. Check convergence or integrate squared outputs on each interval. The reference solution uses exact matrix-exponential propagation and interval energy integrals.

Design tasks and deliverables

Task 1: 30 points | Task 2: 30 points | Task 3: 40 points

Task 1 - PID design (30 points)

- (a) Derive $G_u(p) = S(p)/U(p)$ with $z_r = 0$ and $G_w(p) = S(p)/Z_r(p)$ with $u = 0$. Use the same output sign in both.
- (b) Study the passive response to the 0.2 m road step and the force-to-deflection impulse response with zero road input. State the impulse units and magnitude.
- (c) Tune a PID controller on suspension deflection. Report K_p , K_i , K_d , the feedback sign, and whether derivative filtering is used. MATLAB pidTuner is optional. Its reference-tracking plot is not the road-disturbance response.
- (d) Simulate the closed loop under the specified road inputs and assess the benchmark metrics. Discuss force spikes and practical derivative filtering.

Task 2 - LQR design (30 points)

- (a) Derive the state-space model with $x = [z_s, \dot{z}_s, s, \eta]^T$ using the transformed state defined in the model handout. Identify the actuator and disturbance channels, assess stability, and test controllability.
- (b) Simulate the passive sinusoidal response. Recover physical velocities from the transformed state correctly.
- (c) Augment the model with $\dot{x}_i = s$, choose $Q \succeq 0$ and $R > 0$, and compute an LQR gain. Report Q , R , K and closed-loop poles. Explain the meaning of each state weight.
- (d) Simulate step and sinusoidal road disturbances and compare with the passive and PID cases. Identify each achieved and unmet target.

Task 3 - Analysis and discussion (40 points)

- (a) Explain ride-comfort versus road-holding tradeoffs. (b) Discuss suspension travel, actuator force and practical limits. (c) Assess robustness using evidence, such as varying parameters or disturbances; nominal simulations alone do not establish robustness. (d) Discuss how gain scheduling, data-driven tuning or road preview could help, including data requirements and reliability concerns.

Deliverable and learning outcomes

Submit one PDF containing the model derivation, controller design, code excerpts, plots, comparison tables and discussion. Students should demonstrate mechanical modeling, state interpretation, PID/LQR design, integral augmentation, numerical validation, and clear engineering communication.

Starter: starter/main_file.m. Reference implementation and solution guide: solutions/. MATLAB with Control System Toolbox is required for the controller workflow; pidTuner for an LTI plant is included in Control System Toolbox.